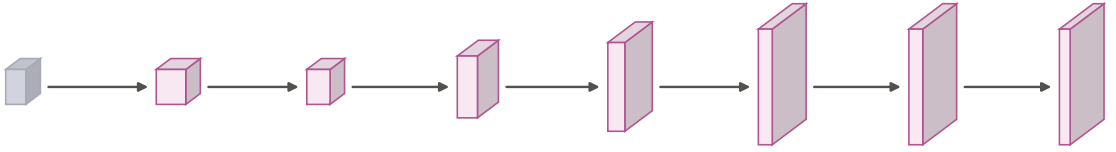
(a) Segmentation decoder: tensor shapes (channels × depth × height × width)



	block tokens (4 taps)	1×1×1 conv, concatenate	3^3 conv	upsample, 3^3 conv	upsample, 3^3 conv	upsample, 3^3 conv	upsample, 3^3 conv	1×1×1 conv
TAP-CT	$768\times3\times28^2$	$4\cdot128\times3\times28^2$	$128\times3\times28^2$	$64\times6\times56^2$	$32\times12\times112^2$	$16\times12\times224^2$	–	$C\times12\times224^2$
SPECTRE	1080×8^3	$4\cdot128\times8^3$	128×8^3	64×16^3	32×32^3	16×64^3	$16\times64\times128^2$	$C\times64\times128^2$

(b) What each segmentation method trains

		early blocks			late blocks					encoder parameters updated	
	patch emb.	W_{qkv}	W_{proj}	MLP, norms	W_{qkv}	W_{proj}	MLP, norms	head	steps	TAP-CT	SPECTRE
Lin								1×1×1 conv	8000	0	0
Dec								decoder	8000	0	0
LoRA								decoder	8000	0.59 M	2.1 M*
LoRA$_{\text{late}}$								decoder	8000	0.20 M	0.69 M
DoRA (+ m)		m			m			decoder	8000	0.62 M	2.2 M*
Dec→LoRA		2			2			decoder	4000 + 4000	0.59 M	2.1 M*
Full FT								decoder	8000	87 M	339 M*

frozen rank-16 adapter (m: DoRA magnitude; 2: added in stage 2) trained in full